

Gross and Applied Anatomy of the Brachial Plexus, Axilla, and Shoulder (MBCHB,BCM)

Courtesy of Dr. M. Evans

19 June, 2025

Introduction

The shoulder region represents one of the most complex and functionally demanding areas of the human body. Its remarkable range of motion, essential for positioning the hand in space, is predicated on an intricate interplay of bony architecture, muscular control, and neurovascular supply. For the medical doctor, a thorough and integrated understanding of this region is not merely an academic exercise but a fundamental prerequisite for accurate diagnosis, safe procedural intervention, and effective management of a wide spectrum of pathologies. This report provides an exhaustive, clinically oriented overview of the **brachial plexus**, the **axilla**, and the **shoulder musculature**, treating them as a cohesive functional unit. The objective is to bridge the gap between foundational gross anatomy and its direct application in clinical practice. The neurovascular network of the brachial plexus, the anatomical conduit of the axilla, and the musculoskeletal machinery of the shoulder will be presented in an interconnected manner. The central theme is the relationship between *structure and function*, and how disruptions in this delicate balance manifest as predictable and recognizable *clinical syndromes*.

Part I

The Brachial Plexus

1 Formation and Topographical Anatomy

The brachial plexus is a sophisticated network of nerves that provides the complete motor and sensory innervation to the upper limb, with the exception of the trapezius muscle (innervated by the spinal accessory nerve, CN XI) and a small patch of skin near the axilla supplied by the intercostobrachial nerve. Its complex organization can be systematically understood using the mnemonic "*Read That Damn Cadaver Book*," which corresponds to its five constituent parts: *Roots, Trunks, Divisions, Cords, and Branches*.

Roots (Ventral Rami of C5-T1)

The brachial plexus originates from the **anterior (ventral) rami** of the lower four cervical spinal nerves (C5, C6, C7, C8) and the first thoracic spinal nerve (T1). It is critical to distinguish these from the **posterior rami**, which course backward to innervate the intrinsic muscles and skin of the back and do not contribute to the plexus. After emerging from their respective **intervertebral foramina**, these five nerve roots course through the neck, passing between the **anterior and middle scalene muscles**. This anatomical passage, known as the **interscalene groove** or triangle, is a clinically significant site for regional anesthesia and is also a potential location for nerve compression.

Trunks (Superior, Middle, Inferior)

As the roots travel laterally, they converge within the **posterior triangle of the neck** to form three distinct trunks. This convergence represents the first intermingling of nerve fibers from different spinal levels:

- The **Superior Trunk** is formed by the union of the C5 and C6 roots.
- The **Middle Trunk** is a direct continuation of the C7 root.
- The **Inferior Trunk** is formed by the union of the C8 and T1 roots.

The trunks traverse laterally, passing over the first rib and the apex of the lung before coursing towards the clavicle.

Divisions (Anterior and Posterior)

Posterior to the middle third of the clavicle, each of the three trunks bifurcates into an **anterior** and a **posterior division**. This yields a total of three anterior and three posterior divisions. This stage is of paramount functional importance as it represents the first major segregation of nerve fibers based on their ultimate destination and function. The *anterior divisions* contain fibers destined to supply the muscles of the anterior (flexor and pronator) compartments of the upper limb, while the *posterior divisions* carry fibers that will innervate the muscles of the posterior (extensor and supinator) compartments. There are no named nerves that branch directly from the divisions.

Cords (Lateral, Medial, Posterior)

After passing behind the clavicle and entering the axilla, the six divisions recombine to form three cords. These cords are named according to their anatomical relationship with the second part of the **axillary artery**, which serves as the central landmark of the axilla.

- The **Lateral Cord** is formed by the union of the anterior divisions of the superior and middle trunks, carrying fibers from C5, C6, and C7.
- The **Medial Cord** is the direct continuation of the anterior division of the inferior trunk, carrying fibers from C8 and T1.
- The **Posterior Cord** is formed by the union of all three posterior divisions, thus containing fibers from all five root levels (C5, C6, C7, C8, T1).

The structure of the plexus is an elegant solution to the complex problem of innervating a limb. The initial convergence of roots into trunks provides *redundancy*, ensuring that most major terminal nerves receive contributions from multiple spinal levels. A lesion affecting a single nerve root, such as C7, will therefore cause weakness but not complete paralysis in the muscles supplied by nerves like the median and radial nerves. The subsequent segregation into anterior and posterior divisions creates a fundamental *functional sorting*. All fibers destined for flexor muscles are bundled together in the anterior divisions, while all extensor fibers are grouped in the posterior divisions. This organization persists into the cords, with the lateral and medial cords being functionally "flexor" and the posterior cord being exclusively "extensor." This hierarchical arrangement simplifies motor control, allowing the central nervous system to activate entire functional muscle groups rather than individual muscles from disparate spinal levels. This anatomical logic is the key to understanding why injuries to different parts of the plexus produce such predictable patterns of functional loss.

Anatomical Variations and Clinical Implications

While the "textbook" description provides a useful framework, significant anatomical variations of the brachial plexus are common, occurring in approximately 50% of individuals. The most well-described are *prefixed* and *postfixed plexuses*. A **prefixed plexus** receives a significant contribution from the C4 ventral ramus, shifting the entire plexus one level superiorly (C4-C8). A **postfixed plexus** receives a contribution from T2, shifting the plexus inferiorly (C6-T2). A postfixed plexus is clinically relevant because the inferior trunk, now containing T2 fibers, is more susceptible to compression by the first rib, which can produce symptoms of *thoracic outlet syndrome*. Beyond root contributions, variations in the formation of trunks, divisions, and cords are also frequently encountered. This anatomical variability has profound implications for clinicians. For surgeons operating in the neck and axilla, and for anesthesiologists performing regional nerve blocks, the assumption of "normal" anatomy can be misleading. An unexpected neurovascular relationship can increase the risk of *iatrogenic nerve injury* or lead to the failure of an anesthetic block.

2 Functional Organization and Branching Pattern

The brachial plexus gives rise to numerous collateral and terminal branches that innervate the muscles and skin of the upper limb. These are best categorized based on their point of origin relative to the clavicle.

Supraclavicular Branches (from Roots and Trunks)

These nerves arise proximal to the clavicle and primarily innervate the scapular and periscapular muscles.

- **Dorsal Scapular Nerve (C5):** Arising directly from the C5 root, it pierces the middle scalene muscle and travels deep to supply the *levator scapulae* and the *rhomboid major and minor* muscles, which are key for scapular elevation and retraction.
- **Long Thoracic Nerve (C5, C6, C7):** Formed from the ventral rami of C5, C6, and C7, this nerve descends along the lateral surface of the *serratus anterior muscle*, providing its sole motor innervation. Its long and superficial course makes it particularly vulnerable to traction injuries, direct trauma, or iatrogenic damage during axillary surgery.
- **Nerve to Subclavius (C5, C6):** A small nerve that arises from the superior trunk and descends to innervate the *subclavius muscle*, which helps stabilize the clavicle.

- **Suprascapular Nerve (C5, C6):** Originating from the superior trunk, it passes through the **suprascapular notch** (inferior to the superior transverse scapular ligament) to supply the *supraspinatus muscle*. It then continues around the **spinoglenoid notch** to innervate the *infraspinatus muscle*. It also provides sensory fibers to the glenohumeral and acromioclavicular joints.

Infraclavicular Branches (from Cords)

These branches arise distal to the clavicle, within the axilla, and include both smaller collateral nerves and the five major terminal nerves of the upper limb.

- **Lateral Cord Branches (C5, C6, C7):** The mnemonic "*Lucy Loves Me*" (LLM) helps recall these branches.
 - **Lateral Pectoral Nerve:** The primary supply to the clavicular head of the *pectoralis major*. It often sends a communicating branch to the medial pectoral nerve.
 - **Musculocutaneous Nerve (Terminal):** This nerve pierces the *coracobrachialis muscle* and provides motor innervation to all muscles in the *anterior compartment of the arm*: coracobrachialis, biceps brachii, and brachialis. Distal to the elbow, it becomes the *lateral cutaneous nerve of the forearm*, providing sensation to the lateral forearm.
 - **Lateral Root of the Median Nerve (Terminal):** This is the lateral contribution to the median nerve.
- **Medial Cord Branches (C8, T1):** The mnemonic "*Most Medical Men Use Morphine*" (MMMUM) is useful here.
 - **Medial Pectoral Nerve:** Supplies the sternocostal head of the *pectoralis major* and the *pectoralis minor muscle*.
 - **Medial Cutaneous Nerve of the Arm:** Provides sensation to the skin on the medial aspect of the arm.
 - **Medial Cutaneous Nerve of the Forearm:** Provides sensation to the skin on the medial aspect of the forearm.
 - **Ulnar Nerve (Terminal):** Descends on the medial side of the arm, passes posteriorly to the medial epicondyle of the humerus (in the *cubital tunnel*), and enters the forearm. It innervates the *flexor carpi ulnaris*, the medial half of the *flexor digitorum profundus*, and most of the intrinsic muscles of the hand.
 - **Medial Root of the Median Nerve (Terminal):** Joins with the lateral root to form the median nerve.
- **Posterior Cord Branches (C5-T1):** The mnemonic "*STAR*" (Subscapular, Thoracodorsal, Axillary, Radial) or "*ULTRA*" can be used.

- **Upper Subscapular Nerve (C5, C6):** Supplies the superior portion of the *subscapularis muscle*.
- **Thoracodorsal Nerve (C6, C7, C8):** Also known as the nerve to *latissimus dorsi*, it runs down the posterior axillary wall to supply this large muscle.
- **Lower Subscapular Nerve (C5, C6):** Supplies the inferior part of the *subscapularis* and the *teres major muscle*.
- **Axillary Nerve (Terminal):** A smaller terminal branch that exits the axilla posteriorly through the **quadrangular space** along with the posterior circumflex humeral artery. It innervates the *deltoid* and *teres minor* muscles and provides sensory innervation to the skin over the deltoid, an area known as the "*regimental badge*".
- **Radial Nerve (Terminal):** The largest branch of the brachial plexus. It travels in the **radial groove** of the humerus and is responsible for innervating all muscles of the *posterior (extensor) compartments of the arm and forearm*.

Part II

The Axilla: Conduit

3 Architectural Anatomy of the Axillary Pyramid

The axilla, or armpit, is far more than a simple space; it is a complex, three-dimensional **pyramidal region** that serves as the primary conduit for all major neurovascular structures passing between the trunk and the upper limb. Its volume and shape are dynamic, changing significantly with movements of the arm, particularly abduction, which narrows the apical inlet and can compress its contents. Understanding its boundaries is essential for any clinical procedure in this region.

- **Apex (Cervicoaxillary Canal):** This superior gateway connects the axilla to the posterior triangle of the neck. It is a truncated triangle bounded by the lateral border of the **first rib** medially, the posterior surface of the **clavicle** anteriorly, and the superior border of the **scapula** posteriorly.
- **Base (Floor):** The floor is formed by the skin of the armpit and the underlying dense **axillary fascia**, which stretches from the anterior to the posterior axillary folds.
- **Anterior Wall:** This powerful wall is formed by the **pectoralis major** muscle superficially and the **pectoralis minor** and **subclavius** muscles deep to it.

- **Posterior Wall:** This wall is composed of muscles that act on the humerus, including the **subscapularis** superiorly, and the **teres major** and **latissimus dorsi** inferiorly.
- **Medial Wall:** This broad wall is formed by the upper four or five ribs and their intercostal muscles, overlaid by the **serratus anterior** muscle.
- **Lateral Wall:** This is a very narrow bony wall, formed by the **intertubercular sulcus** of the humerus.

4 Neurovascular and Lymphatic Contents in Situ

The axilla is densely packed with fat and connective tissue that envelops the vital structures passing through it. These structures are not randomly arranged but are organized within a fascial compartment and in relation to one another.

The Axillary Sheath and its Contents

The **axillary artery**, **axillary vein**, and the cords of the brachial plexus are bundled together within the **axillary sheath**, a sleeve-like fascial extension of the *prevertebral fascia* of the neck. This sheath creates a contained space, which is the precise target for an *axillary brachial plexus block*, allowing local anesthetic to bathe the nerves. The **axillary artery** is the central axis around which the other structures are organized. It is a continuation of the subclavian artery, beginning at the lateral border of the first rib, and becomes the brachial artery as it passes the inferior border of the teres major muscle. The **pectoralis minor muscle** crosses anterior to the artery, dividing it into three clinically relevant parts:

- **First Part (proximal to pectoralis minor):** The lateral and posterior cords of the brachial plexus lie superolateral to the artery, while the medial cord is posterior to it.
- **Second Part (posterior to pectoralis minor):** This is the most critical landmarking zone. The three cords embrace the artery precisely according to their names: the **lateral cord** is lateral, the **medial cord** is medial, and the **posterior cord** is posterior to the artery.
- **Third Part (distal to pectoralis minor):** The cords give rise to their terminal branches, which begin to diverge as they travel down the limb.

The axillary vein lies medial to the axillary artery along its entire course. It receives tributaries corresponding to the branches of the artery and is the main venous drainage channel for the upper limb. The anatomy of the axilla presents a zone of *high-stakes navigation* for clinicians. It is a confined space where the entire neurovascular and lymphatic supply for a limb is concentrated. The **pectoralis minor muscle** serves as an indispensable landmark, dividing the axilla into three distinct zones, each with a different arrangement of the neurovascular bundle. A surgeon performing an *axillary lymph node dissection* for breast cancer must meticulously remove nodal tissue while identifying and preserving the **long thoracic nerve** on the medial wall (to prevent a *winged scapula*) and the **thoracodorsal nerve** on the posterior wall (to preserve latissimus dorsi function). Similarly, an anesthesiologist performing an axillary block relies on the pulsatile axillary artery as a sonographic landmark to locate the surrounding terminal nerves within the axillary sheath. A failure to appreciate these millimeter-scale relationships can lead to procedural failure or catastrophic *iatrogenic injury*.

Axillary Lymph Nodes

Embedded within the axillary fat are 20 to 40 lymph nodes, which are of paramount clinical importance, particularly in the staging and management of **breast cancer**. They are organized into five principal groups that form a hierarchical drainage pathway.

Group Name	Location	Primary Drainage Territory	Efferent Drainage Pathway	Surgical Level
Pectoral (Anterior)	Along the inferior border of pectoralis minor, near the lateral thoracic vessels.	Lateral quadrants of the breast; skin/muscles of anterolateral body wall above umbilicus.	Central and Apical nodes.	Level I
Subscapular (Posterior)	On the posterior axillary wall, along the subscapular vessels.	Skin/muscles of the posterior neck and upper back.	Central and Apical nodes.	Level I
Humeral (Lateral)	Posteromedial to the axillary vein on the lateral axillary wall.	Majority of the upper limb.	Central, Apical, and deep cervical nodes.	Level I

Group Name	Location	Primary Drainage Territory	Efferent Drainage Pathway	Surgical Level
Central	Deep to pectoralis minor, near the base of the axilla, embedded in fat.	Receives lymph from the Pectoral, Subscapular, and Humeral groups.	Apical nodes.	Level II
Apical (Terminal)	At the apex of the axilla, superior to pectoralis minor.	Receives lymph from all other axillary groups and directly from the upper breast.	Subclavian lymphatic trunk.	Level III

Table 1: Principal Groups of Axillary Lymph Nodes

The *surgical levels (I, II, III)* are defined by their relationship to the pectoralis minor muscle, a key landmark in axillary dissection. Level I nodes are lateral to the muscle, Level II are posterior to it, and Level III are medial to it.

Part III

The Shoulder Musculature: Movers and Stabilizers

The shoulder's function is governed by a complex array of muscles that can be broadly categorized into *dynamic stabilizers* and *prime movers*. The **glenohumeral joint**'s design, with a large humeral head articulating with a small, shallow glenoid fossa, creates an inherent *stability-mobility paradox*: its vast range of motion comes at the cost of significant bony instability. Consequently, the shoulder relies heavily on the surrounding musculature for both movement and stability.

5 The Dynamic Stabilizers: The Rotator Cuff (SITS Muscles)

The primary function of the four **rotator cuff muscles** is not to generate powerful movements but to provide dynamic stability to the glenohumeral joint. Their tendons blend with the joint capsule, forming a **musculotendinous cuff** that surrounds the joint superiorly, anteriorly, and posteriorly, leaving a notable area of vulnerability inferiorly. Through coordinated contraction, these muscles compress the humeral head into the glenoid fossa, keeping it centered during motion. The mnemonic *SITS* helps recall the four muscles: Supraspinatus, Infraspinatus, Teres minor, and Subscapularis.

- **Supraspinatus:** Arises from the supraspinous fossa of the scapula and inserts on the superior facet of the greater tubercle of the humerus. It is innervated by the suprascapular nerve (C5, C6). Its primary action is to *initiate the first 0-15 degrees of arm abduction*, after which the deltoid becomes the prime mover.
- **Infraspinatus:** Arises from the infraspinous fossa and inserts on the middle facet of the greater tubercle. It is also innervated by the suprascapular nerve (C5, C6). It is a primary *external (lateral) rotator* of the shoulder.
- **Teres Minor:** Arises from the upper lateral border of the scapula and inserts on the inferior facet of the greater tubercle. It is innervated by the axillary nerve (C5, C6) and assists the infraspinatus in external rotation.
- **Subscapularis:** The only rotator cuff muscle on the anterior aspect of the scapula, it arises from the subscapular fossa and inserts on the lesser tubercle of the humerus. It is innervated by the upper and lower subscapular nerves (C5, C6) and is the primary *internal (medial) rotator* of the shoulder.

6 The Prime Movers and Scapular Anchors

While the rotator cuff provides fine-tuned stability, larger muscles are responsible for powerful movements of the shoulder and for anchoring the scapula to the axial skeleton.

- **Deltoid:** This large, triangular muscle forms the rounded contour of the shoulder. It has three heads with distinct functions: the *anterior (clavicular) head* flexes and medially rotates the arm; the *middle (acromial) head* is the prime mover for arm abduction from 15 to 90 degrees; and the *posterior (spinal) head* extends and laterally rotates the arm. It is innervated by the axillary nerve (C5, C6).

- **Pectoralis Major:** A large fan-shaped muscle on the anterior chest wall. Its clavicular head flexes the arm, while its larger sternocostal head is a powerful adductor and medial rotator of the arm. It is innervated by both the medial (C8, T1) and lateral (C5-C7) pectoral nerves.
- **Latissimus Dorsi:** A broad muscle of the back, it is a powerful extensor, adductor, and medial rotator of the humerus. It is innervated by the thoracodorsal nerve (C6, C7, C8).
- **Teres Major:** Often called the "lat's little helper," it assists in adduction, extension, and medial rotation of the arm. It is innervated by the lower subscapular nerve (C5, C6).
- **Serratus Anterior:** This muscle holds the scapula against the thoracic wall. It is essential for protraction (reaching forward) and upward rotation of the scapula during arm elevation. It is innervated by the long thoracic nerve (C5, C6, C7).
- **Trapezius:** A large superficial muscle of the back and neck. Its upper fibers elevate the scapula, middle fibers retract it, and lower fibers depress it. The upper and lower fibers work together with the serratus anterior to produce upward rotation of the scapula. It is innervated by the spinal accessory nerve (CN XI).
- **Rhomboids (Major and Minor):** Located deep to the trapezius, these muscles retract and downwardly rotate the scapula. They are innervated by the dorsal scapular nerve (C5).

7 The Biomechanics of Shoulder Motion

Healthy shoulder function is a testament to biomechanical harmony, requiring a precise balance of forces and synchronized movements across multiple joints.

Force Couples

Efficient movement and stability are achieved through **force couples**, where two or more muscles on opposite sides of a joint contract simultaneously to produce pure rotation without unwanted translation.

- **Coronal Plane Force Couple:** During arm abduction, the powerful deltoid muscle pulls the humeral head superiorly. To counteract this and prevent the humeral head from jamming into the acromion (impingement), the inferior rotator cuff muscles

(infraspinatus, teres minor, and subscapularis) exert a coordinated downward and inward pull. This creates a force couple that rotates the humeral head on the glenoid while keeping it depressed and centered.

- **Transverse Plane Force Couple:** The anterior pull of the subscapularis is balanced by the posterior pull of the infraspinatus and teres minor. This force couple is essential for keeping the humeral head centered in the glenoid during internal and external rotation.

Scapulohumeral Rhythm

Arm elevation is not solely a glenohumeral joint motion. It involves a coordinated, simultaneous movement of the scapula on the thorax, known as **scapulohumeral rhythm**. For every 3 degrees of arm abduction, approximately 2 degrees occurs at the glenohumeral joint and 1 degree occurs from upward rotation of the **scapulothoracic articulation** (a *2:1 ratio*). This rhythm is critical for two reasons: it increases the total range of motion, and it maintains the glenoid in an optimal position to support the humeral head, thereby preserving the length-tension relationship of the deltoid muscle. This scapular rotation is itself produced by a force couple involving the upper trapezius, lower trapezius, and the serratus anterior muscle. The intricate system of force couples and scapulohumeral rhythm reveals that many common shoulder pathologies are not simply isolated tissue failures but are often the result of a breakdown in this biomechanical harmony. For instance, a tear of the supraspinatus or weakness in the other rotator cuff muscles disrupts the coronal plane force couple. The deltoid's superior pull becomes unopposed, leading to superior migration of the humeral head, which narrows the subacromial space and causes or worsens *impingement syndrome*. Similarly, paralysis of the serratus anterior due to a long thoracic nerve injury disrupts the scapular force couple. The scapula fails to rotate upward correctly during abduction, a condition known as *scapular dyskinesis*. This failure to move the acromion "out of the way" also results in functional subacromial impingement. Therefore, a comprehensive clinical evaluation must assess the entire kinetic chain of the shoulder, not just the site of pain, to identify the root cause of the dysfunction.

Part IV

Clinical Correlates and Applied Medical Science

A deep understanding of the anatomy of the brachial plexus, axilla, and shoulder provides the foundation for diagnosing and treating a wide array of clinical conditions. The predictable patterns of motor and sensory loss following nerve injury, the pathophysiology of common musculoskeletal disorders, and the rationale behind diagnostic and interventional procedures are all rooted in this anatomical

knowledge.

8 Brachial Plexus Plexopathies

Injuries to the brachial plexus, or **plexopathies**, result in deficits that correspond precisely to the level of the lesion. The most classic examples involve the upper and lower trunks.

Upper Trunk Palsy (Erb-Duchenne Palsy)

Anatomical Basis: This is an injury to the superior trunk of the brachial plexus, involving nerve fibers from the C5 and C6 roots. **Mechanism:** The most frequent cause is an excessive increase in the angle between the neck and shoulder, which places the upper trunk under extreme traction. This commonly occurs during a difficult childbirth with *shoulder dystocia* but can also result from falls or trauma in adults where the shoulder is forcefully depressed. **Clinical Presentation:** The injury affects nerves derived from C5-C6, primarily the suprascapular, axillary, and musculocutaneous nerves. This leads to paralysis of the shoulder abductors (supraspinatus, deltoid), external rotators (infraspinatus, teres minor), and elbow flexors (biceps, brachialis). The result is the pathognomonic "*Waiter's Tip*" or "*Porter's Tip*" posture: the arm hangs limply at the side, adducted and internally rotated, with the elbow extended and the forearm pronated. Sensory loss is typically found over the lateral aspect of the arm (C5-C6 dermatomes).

Lower Trunk Palsy (Dejerine-Klumpke Palsy)

Anatomical Basis: This is an injury to the inferior trunk, involving fibers from the C8 and T1 roots. **Mechanism:** The injury mechanism is forceful hyperabduction of the arm, which stretches the lower components of the plexus. Classic examples include grabbing a tree branch to break a fall or a breech delivery where the infant's arm is pulled overhead. **Clinical Presentation:** The C8 and T1 roots provide the primary innervation to the intrinsic muscles of the hand via the median and ulnar nerves. Paralysis of these muscles results in a "*Claw Hand*" deformity, characterized by hyperextension at the metacarpophalangeal (MCP) joints and flexion at the interphalangeal (IP) joints, due to the unopposed action of the long extensors. Sensory loss occurs along the medial aspect of the forearm and hand (C8-T1 dermatomes). **Associated Horner's Syndrome:** The T1 nerve root is in close proximity to the cervical sympathetic chain. A proximal T1 injury can disrupt these sympathetic fibers, leading to an associated **Horner's syndrome** on the ipsilateral side, which includes *ptosis* (drooping eyelid), *miosis* (constricted pupil), and *anhidrosis* (decreased sweating).

Long Thoracic Nerve Palsy

Anatomical Basis: This is an isolated injury to the long thoracic nerve (C5, C6, C7). **Mechanism:** The nerve is highly vulnerable due to its long, superficial course along the lateral chest wall on the surface of the serratus anterior muscle. It can be damaged by traction injuries from repetitive overhead activities, direct blunt trauma, or iatrogenically during axillary lymph node dissection or chest tube insertion. **Clinical Presentation:** The injury causes paralysis of the serratus anterior muscle. This muscle's primary function is to protract the scapula and anchor its medial border to the thoracic wall. Without this anchor, the medial border and inferior angle of the scapula lift away from the back, a sign known as "*Winging of the Scapula*." This is most prominent when the patient performs a push-up or pushes against a wall. This winging disrupts the normal scapulohumeral rhythm, leading to pain and profound difficulty in elevating the arm above the horizontal plane.

Table 2: *
Summary of Brachial Plexus Plexopathies

Syndrome	Level of Injury	Common Mechanism	Key Nerves Affected	Major Muscle Deficits	Cardinal Clinical Sign/Posture
Erb-Duchenne Palsy	C5, C6 (Superior Trunk)	Lateral neck traction (e.g., birth dystocia, fall)	Suprascapular, Axillary, Musculocutaneous	Deltoid, Supraspinatus, Infraspinatus, Teres Minor, Biceps, Brachialis	Arm adducted, internally rotated, elbow extended, forearm pronated (" <i>Waiter's Tip</i> ")
Dejerine-Klumpke Palsy	C8, T1 (Inferior Trunk)	Arm hyperabduction (e.g., grabbing object while falling)	Ulnar, Median (intrinsic hand branches)	Intrinsic muscles of the hand (Lumbricals, Interossei)	" <i>Claw Hand</i> " (MCP hyperextension, IP flexion); possible <i>Horner's Syndrome</i>
Long Thoracic Nerve Palsy	Long Thoracic Nerve (C5, C6, C7)	Traction, compression, iatrogenic injury	Long Thoracic Nerve	Serratus Anterior	" <i>Winging</i> " of the scapula (medial border lifts off thorax) on pushing

9 Anatomical Basis of Common Shoulder Pathologies

Subacromial Impingement Syndrome

Pathophysiology: This syndrome arises from the mechanical compression and irritation of the structures within the **subacromial space**—most notably the supraspinatus tendon and the subacromial bursa. This space is anatomically confined by the humeral head inferiorly and the rigid **coracoacromial arch** (formed by the acromion, coracoacromial ligament, and acromioclavicular joint) superiorly.

Contributing Factors: The pathology can be caused by structural factors that narrow the space, such as a *hooked (Type III) acromion*, or by functional factors that lead to dynamic narrowing. These functional factors are often a result of biomechanical failure: weakness of the rotator cuff muscles allows for unopposed superior migration of the humeral head by the deltoid, while weakness of the scapular stabilizers (e.g., serratus anterior) leads to *scapular dyskinesis* and failure of the acromion to elevate properly during arm abduction.

Rotator Cuff Tears (focus on Supraspinatus)

Mechanism: Tears can be acute, from trauma, or chronic and degenerative, which is more common. Chronic tears are often the endpoint of long-standing impingement syndrome, where repetitive microtrauma leads to tendon fraying and eventual failure. **Anatomical Vulnerability:** The supraspinatus tendon is the most commonly torn rotator cuff tendon, largely due to its anatomical location. It passes directly through the narrow subacromial space, making it susceptible to compression. Furthermore, a region near its insertion on the greater tuberosity has been identified as a "*critical zone*" of relative *hypovascularity*, which may impair its intrinsic healing capacity and predispose it to age-related degeneration and tearing.

Anterior Glenohumeral Dislocation

Anatomical Basis: Accounting for up to 97% of all shoulder dislocations, anterior dislocations are overwhelmingly common due to the inherent anatomical features of the joint: a large, mobile humeral head in a shallow glenoid socket, and a joint capsule that is relatively thin and unsupported at its *anterior-inferior aspect*. **Mechanism:** The classic mechanism involves a force applied to the arm when it is in a position of *abduction and external rotation*, which levers the humeral head out of the front of the glenoid. **Associated Lesions:** The initial traumatic dislocation often creates lasting structural damage that predisposes the joint to future instability.

- **Bankart Lesion:** This is an avulsion of the anterior-inferior **glenoid labrum** from the glenoid rim. The labrum serves as a critical bumper and attachment site for the glenohumeral ligaments; its detachment compromises the static stability of the joint and is a primary reason for recurrent dislocations.

- **Hill-Sachs Lesion:** As the humeral head dislocates anteriorly, its soft, posterolateral aspect impacts against the hard anterior rim of the glenoid, creating a compression fracture or "divot" known as a Hill-Sachs lesion.

Complications:

- **Axillary Nerve Injury:** The axillary nerve wraps intimately around the surgical neck of the humerus as it passes through the quadrangular space. During an anterior dislocation, the nerve is highly susceptible to being stretched or damaged. This occurs in up to 40% of cases and manifests as paralysis or weakness of the deltoid muscle and sensory loss over the *regimental badge area*.
- **Recurrent Instability:** Following a primary traumatic dislocation, particularly in young, active individuals, the risk of recurrence is very high. The initial injury creates permanent anatomical deficits (Bankart lesion, capsular stretching) that reduce the joint's stability, allowing it to dislocate again with progressively less force.

10 Anatomy in Practice: Diagnostic and Interventional Procedures

Clinical practice is, in essence, a series of applied anatomical manipulations. Physical exam maneuvers, regional anesthetic blocks, and surgical approaches are all designed around a precise understanding of the body's structure.

Physical Examination

Provocative tests are not arbitrary movements; they are designed to selectively stress or compress specific anatomical structures to reproduce a patient's symptoms.

- **Hawkins-Kennedy Test:** In this test, the examiner passively flexes the patient's shoulder and elbow to 90 degrees and then internally rotates the arm. This motion directly compresses the greater tuberosity of the humerus against the coracoacromial ligament, narrowing the subacromial space. A positive test, indicated by pain, suggests impingement of the supraspinatus tendon or the subacromial bursa.
- **Jobe's Test (Empty Can Test):** The patient's arm is abducted to 90 degrees in the plane of the scapula (about 30 degrees anterior to the coronal plane) and then fully internally rotated so the thumb points to the floor. The examiner then applies downward pressure. This position isolates the function of the supraspinatus muscle and also brings its tendon into an impingement position. Pain or weakness is considered a positive test, suggestive of supraspinatus tendinopathy or a tear.

Regional Anesthesia

Regional anesthesia is the targeted delivery of medication into a specific fascial compartment to block nerve conduction. The choice of block is a clinical decision based on the required surgical field and the avoidance of potential complications, a decision rooted entirely in anatomy.

- **Interscalene Block:** This block targets the brachial plexus at the level of the roots and trunks as they pass through the interscalene groove. It is ideal for surgery on the shoulder and proximal humerus. Using ultrasound, the clinician identifies the C5, C6, and C7 roots as distinct hypoechoic circles (the "*stoplight*" sign) between the anterior and middle scalene muscles and injects local anesthetic into the surrounding fascial sheath. The primary complication is an almost universal ipsilateral **phrenic nerve palsy**, caused by the inevitable spread of anesthetic to the nearby phrenic nerve, which lies on the anterior surface of the anterior scalene muscle. This is generally well-tolerated in healthy patients but can cause respiratory distress in those with pre-existing pulmonary disease.
- **Axillary Block:** This block targets the terminal branches of the plexus (median, ulnar, radial, and musculocutaneous nerves) deep within the axilla. It is ideal for surgery of the elbow, forearm, and hand. Using ultrasound, the pulsatile axillary artery is located, and the surrounding nerves are identified in their characteristic positions (e.g., median nerve often superior/lateral, ulnar nerve medial, radial nerve posterior). The musculocutaneous nerve, having already exited the axillary sheath to enter the coracobrachialis muscle, must be blocked separately. The major advantage of this block is that it reliably spares the phrenic nerve, making it a safer choice for patients with respiratory compromise.

Surgical Exposures

Surgical approaches are anatomical roadmaps designed to provide access to deep structures while minimizing damage to neurovascular bundles and muscles.

- **Deltopectoral Approach:** This is the workhorse anterior approach to the shoulder, used for procedures like shoulder arthroplasty and complex fracture fixation.
 - **Landmarks and Incision:** The incision follows the **deltopectoral groove**, a line running from the coracoid process towards the deltoid insertion.
 - **Internervous Plane:** The success of this approach hinges on developing the *internervous plane* between the deltoid muscle (innervated by the axillary nerve) and the pectoralis major muscle (innervated by the medial and lateral pectoral nerves). By

dissecting within this plane, both muscles can be retracted without compromising their nerve supply, providing excellent and safe exposure to the underlying proximal humerus and glenohumeral joint.

- **Structures at Risk:** Several structures must be protected. The **cephalic vein** lies within the deltopectoral groove and should be preserved and retracted (typically laterally). Deeper retraction places the **musculocutaneous nerve** at risk medially (as it enters the coracobrachialis 5-8 cm distal to the coracoid) and the **axillary nerve** at risk inferiorly as it courses around the surgical neck of the humerus.

The anatomy of the shoulder region, encompassing the brachial plexus, the axilla, and the surrounding musculature, forms a deeply interconnected functional unit. A clinician's ability to effectively diagnose and treat pathology in this area is directly proportional to their understanding of these intricate relationships.

- **The Brachial Plexus is a Model of Functional Organization:** The systematic arrangement of the plexus—from roots to branches—is not arbitrary but follows a logical pattern of *convergence for redundancy* and *divergence for functional segregation*. The separation into anterior (flexor) and posterior (extensor) divisions is the fundamental organizing principle that dictates the patterns of motor loss seen in clinical plexopathies.
- **The Axilla is a Critical Anatomical Crossroads:** The axilla serves as a tightly packed conduit for the entire neurovascular supply of the upper limb. The consistent spatial relationships of the brachial plexus cords and axillary vein to the axillary artery, particularly in relation to the pectoralis minor muscle, provide an essential roadmap for surgical dissection and regional anesthesia. Deviations from this map, whether due to anatomical variation or pathological distortion, carry significant risk.
- **Shoulder Function is a Product of Biomechanical Harmony:** The shoulder's immense mobility is balanced by a sophisticated system of dynamic muscular stabilization. Pathologies such as rotator cuff tears and impingement syndrome are often not isolated events but rather the endpoint of a breakdown in the delicate balance of *force couples* and *scapulohumeral rhythm*. A successful clinical approach requires an assessment of this entire kinetic chain.
- **Clinical Practice is Applied Anatomy:** From interpreting the "Waiter's Tip" posture of an Erb's palsy to performing a deltopectoral approach, clinical medicine in the shoulder region is a constant application of anatomical principles. The ability to visualize these three-dimensional relationships allows the physician to translate a patient's symptoms into a precise anatomical diagnosis and to execute procedures with efficacy and safety. A mastery of this *applied anatomy* is, therefore, an indispensable tool for any medical doctor managing conditions of the upper limb.